Spam Message Classifier

CSC 425 - Artificial Intelligence
Term Project - Milestone 1 (Project Proposal)
Fall 2026 - Instructor: Dr. Junxiu Zhou

1. Team Members and Responsibilities

I am doing this project on my own. I was not able to join a team, so I am completing the term project as an individual. I have emailed the instructor to make sure this is okay. Because I am working alone, I am responsible for every part of the project. The table below shows how I plan to divide the work across the semester.

Member	Responsibilities
Arjun Lekhak (working alone)	Collect and clean the data, build and train the model, test and improve it, keep the GitHub repository updated, and write the report and presentation.

How I will share and track my work: I will keep all my code and files on GitHub so everything is saved in one place. I will follow the project schedule in Section 7 to stay on time.

2. Problem and Why It Matters

Spam messages are a common problem. They fill up our phones and inboxes with junk, and many of them try to trick people into clicking bad links or giving away personal information. The goal of this project is to build a program that reads a short text message and decides if it is spam (junk) or ham (a normal, real message). This is a good project because it is a clear problem, there is free data available to use, and I can start with a simple model and make it better over time.

3. Input, AI Task, and Output

Input	One text message, for example: "WINNER! Claim your free prize now, click here."
AI Task	Text classification (sorting text into one of two groups using machine learning).
Output	A label that says spam or ham, with a confidence score.

4. Dataset

I will use the SMS Spam Collection dataset from the UCI Machine Learning Repository. It is free and often used for this kind of project.

- **Where it comes from:** UCI Machine Learning Repository (public, free to use).
- **Size:** about 5,574 text messages.
- **What is in it:** each row has two things, the message text and a label (spam or ham).
- **Target (what I predict):** the label, spam or ham. Most messages are ham (about 87%) and fewer are spam (about 13%).

Preprocessing (cleaning the data): Yes, this is needed. I will make all text lowercase, remove punctuation and common words, and turn the words into numbers the model can use (TF-IDF). I will split the data into a training set and a test set so I can check how well the model works.

5. Planned AI Approach

I will start simple and improve step by step, using Python and the scikit-learn library.

- **Step 1 (baseline):** a Naive Bayes model, which is a simple and common choice for spam detection. I will also try Logistic Regression to compare.
- **Step 2 (improve):** adjust settings, add more features, and try other models like a Support Vector Machine (SVM).
- **Step 3 (extra, if time allows):** try a small modern deep-learning model to see if it does better.

How I will measure success: Because most messages are ham, accuracy alone can be misleading. I will mainly look at precision, recall, and F1-score, and I will try to avoid marking real messages as spam by mistake.

6. References and Resources

[1] Almeida, T. A., Gomez Hidalgo, J. M., and Silva, T. P. (2013). Towards SMS Spam Filtering: Results under a New Dataset. International Journal of Information Security Science, 2(1), 1-18. (This is the paper that created the dataset I am using.)

[2] SMS Spam Collection Dataset, UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/228/sms+spam+collection (The dataset I will train and test on.)

[3] scikit-learn: Working With Text Data. https://scikit-learn.org/stable/tutorial/text_analytics/working_with_text_data.html (A step-by-step guide for the exact method I am using.)

[4] scikit-learn: Text feature extraction (TF-IDF). https://scikit-learn.org/stable/modules/feature_extraction.html (Official docs for turning text into numbers.)

[5] NLTK - Natural Language Toolkit. https://www.nltk.org/ (A Python tool for cleaning and preparing text.)

7. Project Timeline

Time Period	What I Will Do
Now - Oct 11	Define the problem, get the data ready, and set up GitHub (this milestone).
Oct 11 - 25	Explore and clean the data, and build a first simple model (baseline).
Oct 26 - Nov 15	Improve the model and try other methods.
Nov 16 - 29	Test the model, look at mistakes, and make it better.
Nov 30 - Dec 11	Finish the report, demo, and presentation.

8. Expected Final Deliverable

At the end I will have a working program that takes a message and says if it is spam or ham, a GitHub repository with all my code and notes, a final written report with my results, and a short demo and presentation.